Customer Shopping Behavior Analysis

Python • PostgreSQL • Power BI

3,900 Transactions | 18 Features | 4 Product Categories

1. Project Overview

An end-to-end analysis of customer shopping behavior using 3,900 retail transactions. The pipeline covers data cleaning in Python, business queries in PostgreSQL, and an interactive Power BI dashboard.

2. Dataset

Attribute	Details
Records	3,900 transactions
Features	18 columns
Missing Values	37 nulls in Review Rating
Categories	Clothing, Accessories, Footwear, Outerwear

Key feature groups: Customer Demographics (Age, Gender, Location, Subscription Status), Purchase Details (Item, Category, Amount, Season, Size, Color), Shopping Behavior (Discount Applied, Previous Purchases, Frequency, Review Rating, Shipping Type, Payment Method).

3. Python — Data Cleaning & EDA

3.1 Load & Explore

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
df.info()
df.describe(include='all')
df.isnull().sum()
```

3.2 Handle Missing Values

37 nulls in Review Rating — imputed using the median per product category:

```
df['Review Rating'] = df.groupby('Category')['Review Rating']
                        .transform(lambda x: x.fillna(x.median()))
```

3.3 Column Standardization

```
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchse_amount'})
```

3.4 Feature Engineering

Age Groups — custom bins defined manually:

```
bins   = [0, 18, 30, 55, 100]
labels = ['Young', 'Adults', 'Middle-Aged', 'Seniors']
df['age_groups'] = pd.cut(df['age'], bins=bins, labels=labels)
```

Purchase Frequency — mapped text labels to numeric days:

```
frequency_mapping = {
    'Fortnightly': 14,  'Weekly': 7,   'Annually': 365,
    'Quarterly': 90,    'Bi-Weekly': 14, 'Monthly': 30,
    'Every 3 Months': 90
}
df['purchase_frequency'] = df['frequency_of_purchases'].map(frequency_mapping)
```

3.5 Redundancy Check & DB Load

```
(df['discount_applied'] == df['promo_code_used']).all()  # True → drop one
df = df.drop(columns='promo_code_used', axis=1)
```

```
from sqlalchemy import create_engine
from urllib.parse import quote_plus
engine = create_engine(

f'postgresql+psycopg2://postgres:{quote_plus(password)}@localhost:5432/customer_shoppin
g_behavior'
)
df.to_sql('customer', engine, if_exists='replace', index=False)
```

4. SQL Business Queries

Ten queries run on PostgreSQL using the cleaned dataset loaded from Python.

Q1. Revenue by Gender

```
SELECT gender, SUM(purchse_amount) AS revenue
FROM customer GROUP BY gender;
```

Gender	Revenue (USD)
Female	75,191
Male	157,890

Insight: Male customers generated ~2.1× more revenue than female customers.

Q2. High-Spending Discount Users

```
SELECT customer_id, purchse_amount, discount_applied FROM customer
WHERE discount_applied = 'Yes'
  AND purchse_amount >= (SELECT AVG(purchse_amount) FROM customer);
```

Metric	Value
Qualifying customers	839
Threshold (~avg spend)	$59.76

Insight: 839 customers used a discount yet still spent above average — discounts attract high-intent buyers.

Q3. Top 5 Products by Review Rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating)::numeric, 2) AS average_rating
FROM customer
GROUP BY item_purchased ORDER BY average_rating DESC LIMIT 5;
```

Rank	Product	Avg Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Gloves lead with a 3.86 rating — ideal for featured placement in campaigns.

Q4. Shipping Type Comparison

```
SELECT shipping_type, ROUND(AVG(purchse_amount), 2)
FROM customer
WHERE shipping_type IN ('Express', 'Standard')
GROUP BY shipping_type;
```

Shipping Type	Avg Purchase (USD)
Standard	58.46
Express	60.48

Insight: Express shipping customers spend ~$2 more on average — they tend to be higher-value buyers.

Q5. Subscribers vs. Non-Subscribers

```
SELECT subscription_status, COUNT(*) AS no_of_customer,
       ROUND(AVG(purchse_amount), 2) AS avg_amount,
       SUM(purchse_amount) AS total_amount
FROM customer GROUP BY subscription_status;
```

Status	Customers	Avg Spend	Total Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Insight: Average spend is nearly identical — the 1,053 subscribers are a concentrated loyalty base to grow.

Q6. Most Discount-Dependent Products

```
SELECT item_purchased,
       ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) /
COUNT(*), 2)
       AS discountapplied
FROM customer GROUP BY item_purchased
ORDER BY discountapplied DESC LIMIT 5;
```

Rank	Product	Discount Rate (%)
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Insight: Half of all Hat purchases use discounts — margin impact should be reviewed for these products.

Q7. Customer Segmentation

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1             THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segement
  FROM customer
)
SELECT customer_segement, COUNT(*) AS no_of_customer
FROM customer_type GROUP BY customer_segement ORDER BY no_of_customer DESC;
```

Segment	Customers	% of Total
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Insight: 79.9% of customers are Loyal — strong retention. Focus on converting Returning (18%) to Loyal.

Q8. Top 3 Products per Category

```
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(*) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(*)) AS item_rank
  FROM customer GROUP BY category, item_purchased
)
```

```
SELECT * FROM item_counts WHERE item_rank <= 3;
```

Category	Rank	Product	Orders
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Accessories	3	Belt	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Q9. Repeat Buyers & Subscriptions

```
SELECT subscription_status, COUNT(*) AS no_of_customer
FROM customer WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Subscription Status	Repeat Buyers (>5 purchases)
No	2,518
Yes	958

Insight: 72% of repeat buyers are not subscribed — a clear opportunity to pitch subscription benefits.

Q10. Revenue by Age Group

Age groups defined in Python using custom bins: [0–18] = Young, [18–30] = Adults, [30–55] = Middle-Aged, [55–100] = Seniors.

```
SELECT age_groups, SUM(purchse_amount) AS revenue
FROM customer GROUP BY age_groups ORDER BY revenue DESC;
```

Age Group	Age Range	Total Revenue (USD)
Adults	18 – 30	~62,000
Middle-Aged	30 – 55	~118,000
Seniors	55 – 100	~53,000
Young	0 – 18	~2,000

Insight: Middle-Aged (30–55) dominates revenue due to the widest age bracket. Adults (18–30) are the next strongest segment per capita.

5. Power BI Dashboard

KPI	Value
Total Customers	3,900
Avg Purchase Amount	$59.76
Avg Review Rating	3.75 / 5.00
Subscription Rate	27%

Slicers

- Subscription Status, Gender, Category, Shipping Type

Charts

- Donut — % Customers by Subscription Status
- Bar — Revenue by Category
- Bar — Sales Volume by Category
- Bar — Revenue by Age Group
- Bar — Sales by Age Group

6. Key Findings

- Male customers generate ~2.1× more revenue than female customers.
- 79.9% of customers are Loyal — retention is strong.
- Only 27% subscribe, yet average spend is identical to non-subscribers.
- 72% of repeat buyers are not subscribed — biggest conversion opportunity.
- Middle-Aged (30–55) leads total revenue due to bracket size.
- Hat has a 50% discount rate — highest among all products.
- Gloves (3.86), Sandals (3.84), and Boots (3.82) are top-rated products.

7. Business Recommendations

Priority	Action	Reason
High	Run subscription drive for repeat buyers	72% of repeat buyers unsubscribed
High	Promote Gloves, Sandals, Boots in campaigns	Top-rated → strong social proof
Medium	Review discount strategy for Hat & Sneakers	~50% discount rate hurts margins
Medium	Loyalty rewards for Returning customers	Convert 18% Returning to Loyal
Low	Target Adults (18–30) with mobile-first content	Second-largest revenue segment

8. Repository Structure

```
customer-shopping-behavior-analysis/
├── data/
│   └── customer_shopping_behavior.csv
├── notebooks/
│   └── customer_shopping.ipynb
├── sql/
│   └── customer_shopping_sql1.sql
├── dashboard/
│   └── customer_shopping.pbix
└── README.md
```

9. How to Run

1 — Clone

```
git clone https://github.com/<your-username>/customer-shopping-behavior-analysis.git
```

2 — Install dependencies

```
pip install pandas sqlalchemy psycopg2-binary
```

3 — Run the notebook

```
jupyter notebook notebooks/customer_shopping.ipynb
```

Update the DB password in the notebook's connection string before running.

4 — Run SQL queries

```
psql -U postgres -d customer_shopping_behavior -f sql/customer_shopping_sql1.sql
```

5 — Open dashboard

Open customer_shopping.pbix in Power BI Desktop and refresh the data source connection.

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